CNVgears: an R Package to Integrate and Analyse CNV Calling Results from Multiple Methods and Data Types in a Uniformed and Standardized Manner

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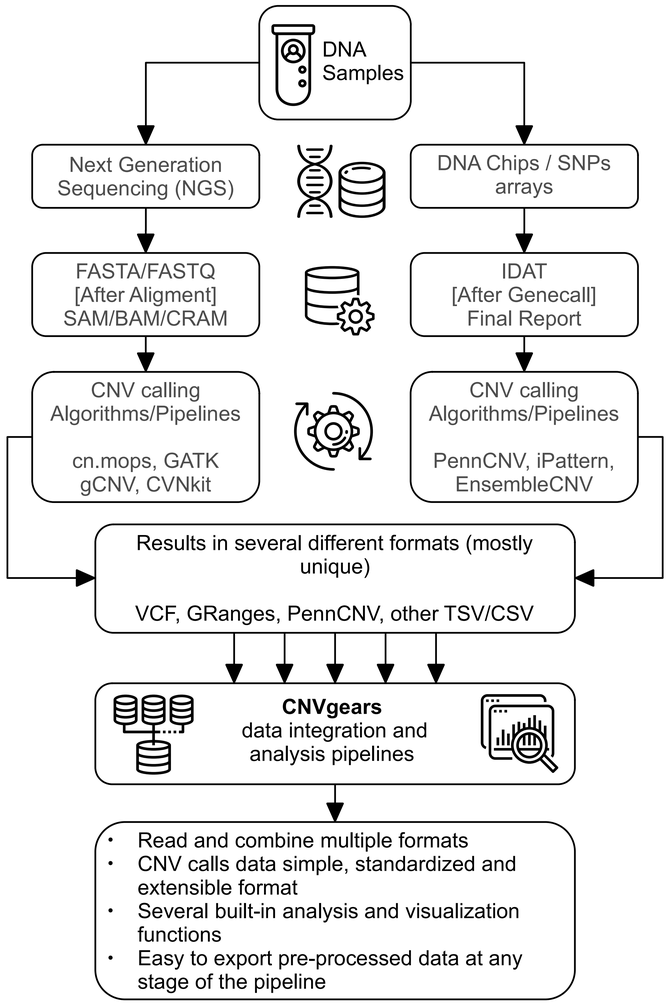
Project aims

CNVgears aims to easily **integrate** results from multiple CNV calling methods, enabling the user to spend less time combining different formats and more **interpreting** the actual results. Several **analysis** can be made with functions provided in the package. We also propose a minimal extensible standard **format** for CNV calling results. We have recently started the Bioconductor submission

CNV calling schematics



Icons sourced from the Noun Project (https://thenounproject.com/) users: Kid Kitaro, Juicy Fish, Becris and monkik.

Introduction

Copy Number Variation (CNVs) are a form of genomic variation known to be very relevant in several psychiatric and neurodevelopmental diseases, such as schizophrenia and autism spectrum disorder (ASD)[1]. Several algorithms exist in order to perform CNV calling, from the "classic" based on CBS[2] (Circular Binary Segmentation) and HMM[3,4] (Hidden Markov Model) to more recent GMM[5] (Gaussian Mixture Model) and HHMM[6] (Hierarchical HMM) based. Moreover, pipelines using more methods exist[7]. However, a clear consensus on how to interpret the results of those calling methods and, more importantly, a uniformed method to integrate, analyze and store those results, regardless of the algorithms and data type used, are still missing.

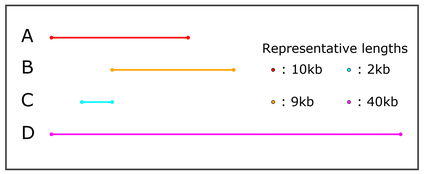
Selected Methods

Here are highlighted the three most important technical part of the package, namely:

1. The standard minimal CNVs format;
2. The reciprocal overlap rule;

chr	start	end	GT	sample_ID
<int>	<int>	<int>	<int>	<char>
1-24	any	any	0/1/2	any

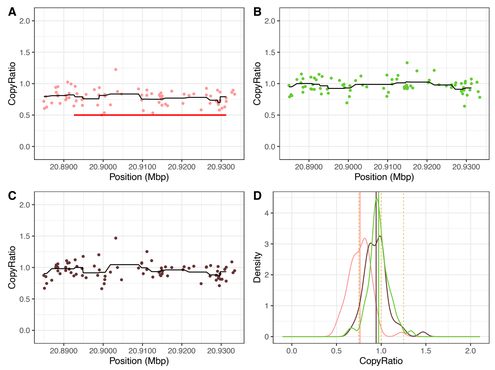
The format we use contains all the minimal informations to define a genomic segment or a CNV in a cohort study, i.e. genomic location (chr, start, end), genotype (0 for normal copy number, 1 for deletions and 2 for duplications) and sample ID. Similarly to the BED format, it is extensible with several more columns available, the easier examples being the precise copy number (CN) or number of markers points (NP).



30% reciprocal overlap is the default rule we use whenever is needed to compare two events in order to determine if they can be the same call. We choose reciprocal over simple overlap because it requires also a similarity in length, A and B can be merged but A and C or A and D can not.

Numeric relationships between Log2R, copyratio and numeric Copy Number (for human somatic chromosomes):

$$LRR = log2R = log2(CR) = log2(numeric_CN/2)$$



Markers-level screening can confirm a putative de-novo call, i.e. a CNV in the offspring in a locus where no parents have events. The points LRR (Log2 R Ratio) or CR (Copy Ratio) means can be compared in the trio (offspring vs mother & offspring vs father). Here we see a good de-novo events in red and the parents in green and brown, the offspring CR mean is statistically significant lower than the parent

CNVgears pipeline schematics

Data load

Individual methods results, intervals/SNPs genomic location, cohort information (PED), markers-level raw data. Summary statistics on the cohort and the results can be easily produced and analyzed for an efficient filtering process.

Filtering & Integration

Filtering can be made using several parameters, then individual results are combined, merging compatible calls within each sample (30% reciprocal overlap).

Processing

Several results-processing function are available including inheritance patterns and Copy Number Variable Regions (CNVRs) detection.

Annotation

Basic annotation functions are already available such as genomic locus and genic content extraction. Subsets of CNV can be selected, for example rare and/or de novo, in a specific locus or detected by more than one method.

Visual confirmation & Data export

Particularly relevant findings can be visually validated using raw data trends. Finally, results can be easily exported in various formats.

Icons sourced from the Noun Project (https://thenounproject.com/) users: Becris, Juicy Fish and mitochondrial.

References

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CNVgears GitHub repository, https://github.com/SinomeM/CNVgears

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